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Final Report

Title: A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)

Calvert Study No.: 0875SL35.001

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA 92121

Date: 9 Dec 2014

I. Table of Contents

II. List of Abbreviations

%	Percent
%RSD	Percent relative standard deviation
°C	Degrees Celsius
µg	Microgram
µg/mL	Microgram per milliliter
µg*hr/mL	Microgram x hour per milliliter
$AUC_{0\text{-}last}$	Area under the curve from time zero to the time of the last quantifiable concentration
$AUC_{0\text{-}inf}$	Area under the curve from time zero to infinity
BLQ	Below the limit of quantitation
C_o	Extrapolated concentration at time zero
C_{last}	Last quantifiable concentration
Cl	Apparent plasma clearance
C_{max}	Maximum observed concentration; for intravenous bolus dosing the C_{max} is considered to occur at the first sampling time after dose administration
Conc	Concentration
g	Gram
hr	Hour
hr:min:sec	Hours:minutes:seconds
IACUC	Institutional animal care and use committee
IV	Intravenous
K_2EDTA	Di-potassium ethylenediaminetetraacetic acid
kg	Kilogram
M	Male
mg	Milligram

mg/kg	Milligram per kilogram
mg/mL	Milligram per milliliter
min	Minute
mL	Milliliter
mL/kg	Milliliter per kilogram
NA	Not applicable
ND	Not determined
No.	Number
NRC	National Research Council
rpm	Revolutions per minute
SD	Standard deviation
sec	Second
SOP	Standard Operating Procedure
$T_{1/2}$	Apparent plasma elimination phase half-life
T_{last}	Time of last quantifiable concentration
T_{max}	Time when the maximum concentration was observed; for intravenous bolus dosing the C_{max} is considered to occur at the first sampling time after dose administration
Trt	Treatment
V_c	Apparent volume of distribution from the central compartment
V_{ss}	Apparent volume of distribution at steady-state
Vol	Volume

III. Signatures

Calvert Study No.: 0875SL35.001

Title: A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)

A. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained.

William G. Tuman   9 Dec 2014

William G. Tuman, M.S.   Date
Study Director
Calvert Laboratories, Inc.

B. Signature of Other Responsible Personnel

Scientific Oversight and Report Review

9 DEC 2014

Scientific Management   Date
Calvert Laboratories, Inc.

IV. Summary

A. Title

A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)

B. Objective

The purpose of this study was to evaluate the pharmacokinetics of LT3114 in male non-naïve Cynomolgus monkeys after a single intravenous dose of LT3114 at two different dose levels.

C. Methods

One group of three male Cynomolgus monkeys was administered LT3114 as a single manual slow bolus intravenous injection over a period of ~3 minutes into a saphenous vein. Following a 29 day washout interval, the same group of monkeys was administered another intravenous dose of LT3114. Targeted protocol specified parameters for each treatment are summarized in the table below.

Trt	# of Animals and Sex	Route of Administration	Dose Conc. (mg/mL)	Dose Volume (mL/kg)	Dose Level (mg/kg)
1	3 males[a]	IV	2	5	10
2	3 males[a]	IV	10	5	50

[a]The same three animals were used in each treatment; treatments were separated by a 29 day washout/recovery period.

Following the dose administration for each treatment, serial blood samples were obtained from each animal at 0, 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose. Derived plasma samples collected for each treatment were stored frozen at -80°C until shipment to Burleson Research Technologies (Morrisville, NC). LT3114 monkey plasma concentration assay results were forwarded to Calvert for pharmacokinetic analysis.

D. Results

Intravenous administration via a 3-minute slow bolus hand-push injection was conducted in all three animals per treatment without incident. All blood samples were harvested at their targeted time of collection. All animals appeared normal during dosing and after dosing. Mean assayed LT3114 concentrations for the dosing formulations were 2.2 ± 0.12 mg/mL (%CV = 5.2) and 10.2 ± 0.50 mg/mL (%CV = 4.9).

Nominal mean ± standard deviation dosing details are summarized in the following table.

Trt	Animals and Sex	Route of Administration	Nominal Dose Conc. (mg/mL)	Nominal Dose Volume (g/kg)	Nominal Dose Level (mg/kg)
1	1-3 (male)	IV	1.999	5.27 ± 0.143	10.5 ± 0.29
2	1-3 (male)	IV	10.001	5.15 ± 0.108	51.5 ± 1.08

Mean ± SD; N=3

The nominal dose concentration was based on the actual formulation procedure. For Treatment 1, 4.36 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 65.64 mL of vehicle (nominal dose concentration of 1.999 mg/mL). For Treatment 2, 21.81 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 48.19 mL of vehicle (nominal dose concentration of 10.001 mg/mL). The nominal dose volume was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle. The nominal dose level was calculated as the nominal dose concentration multiplied by the nominal dose volume.

Using noncompartmental analysis and WinNonlin Version 5.3 software, pharmacokinetic parameters were calculated for each individual animal using the animal's LT3114 plasma concentration-time data. Individual animal nominal dose level based on the animal's body weight and nominal dose concentration and nominal dose volume was used in the pharmacokinetic analysis. Mean ± standard deviation pharmacokinetic parameter values were then calculated from the three animals in each treatment group.

A summary table of the primary pharmacokinetic parameters is presented below.

Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{last} (µg/mL)	T_{last} (hr)	AUC_{0-last} (µg*hr/mL)	AUC_{0-inf} (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	1	10.8	235	9.10	480	43301	44978	89.7	0.240	43.8	46.2
1	2	10.5	277	50.1	480	44908	68851	325	0.153	69.3	38.0
1	3	10.3	341	48.7	480	42029	62596	303	0.165	70.3	30.1
	Mean	10.5	284	36.0	480	43413	58808	239.2	0.186	61.1	38.1
	SD	0.29	53.7	23.28	0.0	1442.7	12379.0	130.0	0.04744	15.01	8.06
	%RSD	2.7	18.9	64.7	0.0	3.3	21.0	54.3	25.5	24.6	21.2

Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{last} (µg/mL)	T_{last} (hr)	AUC_{0-last} (µg*hr/mL)	AUC_{0-inf} (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
2	1	50.3	1292	0.0423	180	57640	57641	9.58	0.873	34.9	39.0
2	2	51.8	1340	300	480	251235	393377	321	0.132	61.6	38.7
2	3	52.4	1532	322	480	255554	446165	405	0.117	66.6	34.2
	Mean	51.5	1388	207	380	188143	299061	245	0.374	54.4	37.3
	SD	1.08	127.0	179.9	173.2	113039.5	210735.3	208.3	0.43195	17.04	2.65
	%RSD	2.1	9.1	86.7	45.6	60.1	70.5	85.0	115.5	31.3	7.1

Based on C_0, AUC_{0-last}, AUC_{0-inf} and concentration-time profiles, exposure to LT3114 was LT3114 dose-dependent (in all three animals) and this appeared to be dose-proportional (in 2 of 3 animals). Mean C_0 was 284 ± 53.7 µg/mL and 1388 ± 127.0 µg/mL for nominal mean 10.5 mg/kg and nominal mean 51.5 mg/kg doses, respectively. Monkey #1 was observed with less exposure to LT3114 and greater elimination and clearance of LT3114 when compared to the other two monkeys in both treatments. AUC_{0-inf} ranged from 62596 to 68851µg*hr/mL for two monkeys at the low dose and AUC_{0-inf} ranged from 393377 to 446165 µg*hr/mL for the same two monkeys at the high dose; for monkey #1, AUC_{0-inf} values were 44978 and 57641 µg*hr/mL. Terminal plasma $T_{1/2}$ ranged from 303 to 405 hours for two monkeys; for monkey #1, $T_{1/2}$ values were 89.7 and 9.58 hours following the low and high dose, respectively. Systemic clearance and apparent volume of distribution values were similar per dose level for monkey #2 and #3; Cl values ranged from 0.117 to 0.165 mL/hr/kg and V_{ss} values ranged from 61.6 to 70.3 mL/kg. For monkey #1, the Cl was about 2-7 times greater and its V_{ss} was about two times less. Mean V_c was ~37-38 mL/kg.

V. General Information

A. Key Study Dates

Study Initiation Date:	29 Jul 2014
Experimental Start Date (in-life):	27 Aug 2014
Experimental Completion Date (in-life):	21 Oct 2014

B. Responsible Personnel at Testing Facility

Study Director:	William G. Tuman, B.S.
Primary Technicians:	Melissa Cicchella, B.S., Carrie Olechna, A.S., Erin Stincavage, B.S., Jennifer Shinese, A.S., ALAT and Jennifer Jenson, B.S.

C. Contributing Scientist

Study Monitor:	Marlene W. Modi, Ph.D. MWM Consulting Group Inc. for Lpath Inc. 14 Joseph Court Monmouth Junction, NJ 08852
Sponsor Representative:	Gary Woodnutt, Ph.D. Senior Vice President Research, Lpath, Inc. 4025 Sorrento Valley Blvd. San Diego, CA 92121
Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis:	Victor J. Johnson, Ph.D. Burleson Research Technologies 120 First Flight Lane Morrisville, NC 27560

D. Objective

The purpose of this study was to evaluate the pharmacokinetics of LT3114 in male naïve Cynomolgus monkeys after a single intravenous dose of LT3114 at two different dose levels.

VI. Materials and Methods

A. Test Article

1. Test Article

Identification:	LT3114
Lot/Batch No. :	103948
Physical Description:	Clear colorless liquid
Storage Conditions:	Stored refrigerated (2-8°C) and protected from light.

B. Dose Preparation

Before dose preparation, 2 mL of a 10% polysorbate 80 solution (provided by the Sponsor; Batch# 104012-001, MM#306191), was added aseptically to each liter of formulation buffer (also supplied by the Sponsor and in sterile containers; Batch# 104011-001, MM#306190), and mixed gently to ensure distribution throughout the volume of buffer. The Calvert batch number for this vehicle/diluent was 30 June 2014 and was stored at 2 to 8 °C until use. The Sponsor provided the test article already formulated at 32.1 mg/mL in sterile containers (a Certificate of Analysis for this formulated material is included in Appendix I). Vehicle (24.3 mM citrate, 51.4 nM sodium phosphate, and 300 mM D-trehalose containing the 2 mL of a 10% polysorbate 80 solution) and test article were removed from refrigeration and allowed to warm to room temperature for ~40 minutes. The targeted 2 mg/mL LT3114 formulation was prepared on 27 Aug 2014 and the targeted 10 mg/mL LT3114 formulation was prepared on 24 Sep 2014. All formulation procedures on each day were conducted in a cleaned laminar flow hood using aseptic techniques.

<u>Targeted 2 mg/mL LT3114 Preparation:</u> 4.36 mL of the nominal 32.1 mg/mL LT3114 test article solution was mixed with 65.64 mL of the vehicle. The contents of the formulation vial were gently swirled until the preparation appeared homogeneous. Triplicate 1.0 mL samples were collected and stored at 2 to 8 °C. pH was measured at 5. The nominal LT3114 concentration was 1.999 mg/mL.

<u>Targeted 10 mg/mL LT3114 Preparation:</u> 21.81 mL of the nominal 32.1 mg/mL LT3114 test article solution was mixed with 48.19 mL of the vehicle. The

contents of the formulation vial were gently swirled until the preparation appeared homogeneous. Triplicate 1.0 mL samples were collected and stored at 2 to 8 °C. pH was measured at 5. The nominal LT3114 concentration was 10.001 mg/mL.

C. Dose Formulation Samples

Dose formulation samples were shipped refrigerated on chill packs on 16 Sep 2014 (Treatment 1) and 15 Oct 2014 (Treatment 2) and analytical evaluation was done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. A cover letter and sample transmittal form was sent along with the samples.

D. Accountability and Disposition

Unused test article will be returned to the Sponsor at the termination of this study or, if requested by the Sponsor, retained for use on related future studies.

E. Test System (Animals and Animal Care)

1. Description

Species:	Monkey
Stock:	Cynomolgus (Macaca fascicularis)
Total Number:	3
Gender:	Male
Age Range:	~4 to 4.5 years at start of dosing; records of dates of birth for animals used in this study are retained in the Calvert archives.
Body Weight Range:	3.1 to 3.5 kg at start of dosing
Animal Source:	SNBL (Texas); Cambodia origin
Experimental History:	Purpose-bred and experimentally naïve at the outset of the study.
Identification:	Chest tattoo and cage card

2. *Rationale for Choice of Species and Number of Animals*

The cynomolgus monkey was selected as the non-rodent species used in toxicology studies based upon the substantial amounts of published historical data (1). The test article is a humanized antibody. Nonclinical studies designed to test humanized antibodies are generally conducted in nonhuman primates rather than other large animal species because the antibodies are likely to be reactive in nonhuman primates (2). No previous research has been conducted with the test article in nonhuman primates. The same three animals were used per dose level in crossover fashion to meet the minimum numbers for descriptive statistical analysis of the data.

3. *Husbandry*

Housing:	Animals were housed during acclimation in compliance with USDA Guidelines (7 CFR 2.22, 2.80 and 371.2) and Calvert SOP VET-44. Animals were housed accordingly during the in-life portion of the study. The room in which the animals were kept was documented in the study records. No other species were kept in the same room.
Lighting:	12 hours light/12 hours dark except when room lights were turned on during the dark cycle due to in-life procedures.
Room Temperature:	22 to 25°C
Relative Humidity:	45 to 70%
Food:	All animals had access to Harlan Teklad Primate Diet (certified) or equivalent approximately 4% of their body weight daily. Chemical analysis of the feed was performed by the manufacturer. Results of these analyses are kept on file at Calvert. Certified Primatreats®, fruit and food suitable for human consumption were also given as a supplement at least once daily. The lot number(s) and specifications of each lot of Primatreats® and

	Certified Primate Diet (Teklad) used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which are maintained in the Calvert archives.
Water:	Water was available *ad libitum* to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants were known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives.
Acclimation:	Study animals were acclimated to their housing for 12 months.

4. Prestudy Health Screen and Selection Criteria

All animals received for this study were assessed as to their general health by a member of the veterinary staff or other authorized personnel. During the acclimation period, each animal was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were determined to be suitable for use were assigned to this study.

5. Assignment to Study Groups

Because of the small number on study, the monkeys were manually assigned from an appropriate pool of animals.

6. Humane Care of Animals

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert IACUC approved the study protocol prior to finalization.

F. Test Article Administration

1. Treatment and Dosing Details

Trt	# of Animals and Sex	Route of Administration	Dose Conc. (mg/mL)	Dose Volume (mL/kg)	Dose Level (mg/kg)
1	3 males[a]	IV	2	5	10
2	3 males[a]	IV	10	5	50

[a]The same three animals were used in each treatment; treatments were separated by a 29 day washout/recovery period. The dose concentration, dose volume, and dose level values in the table above represent the protocol targeted values.

2. Dosing

Route: Intravenous

Frequency: Single bolus intravenous administration per treatment.

Procedures: A single intravenous dose was administered into a saphenous vein per treatment. Each intravenous dose was administered as a manual slow bolus injection over a period of approximately 3 minutes.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans. Dose levels were selected by the Sponsor based upon the findings of a dose range finding study previously conducted by the Sponsor. The lowest pharmacologically active dose is 30 mg/kg.

G. In-Life Observations and Measurements

1. Clinical Observations

Animals were observed at least once daily and at all times of blood sampling. Any abnormal clinical signs were recorded.

2. Body Weight

Body weight was determined on each day of test article administration just prior to dosing.

H. Pharmacokinetics

1. Dosing Syringes

The actual dose administered to the animal was the difference between the loaded and unloaded dosing syringe weights. Syringe weights were measured and recorded to two decimal places.

2. Blood Sampling

Following each dose administration, serial blood samples (~1 ml each) were collected by venipuncture of a femoral vein prior to dosing (0 hour) and at 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose. Blood samples were collected into chilled tubes containing K_2EDTA as the anti-coagulant. Blood tubes were placed on chill packs until being centrifuged within 20 minutes of collection.

3. Plasma Harvesting

Blood samples were centrifuged at 3000 rpm at +4 °C for 10 minutes. Each derived plasma sample was transferred into a pre-labeled plastic tube labeled with the study number, species, animal number, route, dose level/test article, date of collection, time of collection, and sample type. Derived plasma samples were stored frozen at approximately -80 °C until shipment to Burleson Research Technologies.

I. Terminal Procedures

1. Disposition of Study Animals

Following the final blood collection for Treatment 2, the monkeys were transferred back into Calvert's in-house primate colony.

J. Sample Shipment

Plasma samples were shipped frozen on dry ice via overnight courier on 16 Sep 2014 (Treatment 1) and 15 Oct 2014 (Treatment 2) to Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. A cover letter and sample transmittal form and sample log was sent along with each set of the samples. Sample receipt by Burleson Research Technologies was confirmed on 17 Sep 2014 (Treatment 1) and 16 Oct 2014 (Treatment 2).

VII. Records and Reports

A. Data Analysis and Statistical Evaluation

Data analysis by Calvert consisted of descriptive statistical analysis of animal body weights and dosing details. Animal observations and blood sampling times (actual and elapsed) were documented. Analytical analysis of the dosing formulations was the responsibility of Burleson Research Technologies. Bioanalysis of the plasma samples was the responsibility of Burleson Research Technologies (BRT). Pharmacokinetic analysis of the plasma sample assay results was the responsibility of Calvert. Individual animal concentration-time data and averaged concentration-time data per dose level were used and Office Excel 2010 (Microsoft Corp., Redmond, WA) was used for graphing and figures. Any plasma value reported as BLQ was assigned a value of zero. Using noncompartmental analysis and WinNonlin Version 5.3 software, individual pharmacokinetics parameters were calculated. These included $AUC_{0\text{-inf}}$, $AUC_{0\text{-last}}$, C_{max}, Cl, C_{last}, C_0, T_{last}, T_{max}, $T_{1/2}$, V_{ss} and V_c. $T_{1/2}$ was determined from regression analysis of the last four consecutive timepoints with quantifiable LT3114 concentration. V_c was calculated by Dose / C_0. Non-truncated values were used in the calculation of all pharmacokinetic parameters and for all descriptive statistical analysis.

B. Storage of Records

Test article tracking, in-life data, protocol, protocol amendment and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. Approximately 2 years after finalization of the report, the Sponsor will be contacted to determine final disposition of all study materials.

VIII. Results

Individual and mean animal body weights and dosing details are summarized in Table 1. Animal observations and blood sampling times (actual and elapsed) are summarized in Appendix III. Individual animal LT3114 concentration-time data are presented in Appendix IV. A BRT analytical/bioanalytical subreport entitled "Determination of LT3114 in Dose Formulation Samples and Monkey Plasma Pharmacokinetic Samples" is presented in Appendix V. Individual and mean ± SD (%RSD) animal LT3114 concentration-time data are summarized in Tables 2

(Treatment 1) and 3 (Treatment 2). Individual LT3114 plasma concentration-time data from both treatments are presented in Figure 1. Mean ± SD LT3114 plasma concentration-time data from each treatment are presented in Figure 2. Individual and mean ± SD (%RSD) LT3114 plasma pharmacokinetic parameters per treatment are summarized in Table 4.

Intravenous administration via a 3-minute slow bolus hand-push injection was conducted in all three animals per treatment without incident. All blood samples were harvested at their targeted time of collection. All animals appeared normal during dosing and after dosing. Mean assayed LT3114 concentrations for the dosing formulations were 2.2 ± 0.12 mg/mL with triplicate middle sample LT3114 measured concentrations of 2.1, 2.3, and 2.3 mg/mL (%CV = 5.2) and 10.2 ± 0.50 with triplicate middle sample LT3114 measured concentrations of 10.7, 9.7, and 10.2 mg/mL (%CV = 4.9). Assayed LT3114 concentration values were not used in the pharmacokinetic analysis.

Nominal mean ± standard deviation dosing details are summarized in the following table.

Trt	Animals and Sex	Route of Administration	Nominal Dose Conc. (mg/mL)	Nominal Dose Volume (g/kg)	Nominal Dose Level (mg/kg)
1	1-3 (male)	IV	1.999	5.27 ± 0.143	10.5 ± 0.29
2	1-3 (male)	IV	10.001	5.15 ± 0.108	51.5 ± 1.08

Mean ± SD; N=3

The nominal dose concentration was based on the actual formulation procedure. For Treatment 1, 4.36 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 65.64 mL of vehicle (nominal dose concentration of 1.999 mg/mL). For Treatment 2, 21.81 mL of the 32.1 mg/mL LT3114 test article solution was mixed with 48.19 mL of vehicle (nominal dose concentration of 10.001 mg/mL). The nominal dose volume was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle. The nominal dose level was calculated as the nominal dose concentration multiplied by the nominal dose volume.

Using noncompartmental analysis and WinNonlin Version 5.3 software, pharmacokinetic parameters were calculated for each individual animal using the animal's LT3114 plasma concentration-time data. Individual animal nominal dose level based on the animal's body weight and nominal dose concentration and nominal dose volume was used in the pharmacokinetic analysis. Mean ± standard deviation pharmacokinetic parameter values were then calculated from the three animals in each treatment group.

A summary table of the primary pharmacokinetic parameters is presented below.

Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	1	10.8	235	9.10	480	43301	44978	89.7	0.240	43.8	46.2
1	2	10.5	277	50.1	480	44908	68851	325	0.153	69.3	38.0
1	3	10.3	341	48.7	480	42029	62596	303	0.165	70.3	30.1
	Mean	10.5	284	36.0	480	43413	58808	239.2	0.186	61.1	38.1
	SD	0.29	53.7	23.28	0.0	1442.7	12379.0	130.0	0.04744	15.01	8.06
	%RSD	2.7	18.9	64.7	0.0	3.3	21.0	54.3	25.5	24.6	21.2

Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
2	1	50.3	1292	0.0423	180	57640	57641	9.58	0.873	34.9	39.0
2	2	51.8	1340	300	480	251235	393377	321	0.132	61.6	38.7
2	3	52.4	1532	322	480	255554	446165	405	0.117	66.6	34.2
	Mean	51.5	1388	207	380	188143	299061	245	0.374	54.4	37.3
	SD	1.08	127.0	179.9	173.2	113039.5	210735.3	208.3	0.43195	17.04	2.65
	%RSD	2.1	9.1	86.7	45.6	60.1	70.5	85.0	115.5	31.3	7.1

Based on C_0, $AUC_{0\text{-}last}$, $AUC_{0\text{-}inf}$ and concentration-time profiles, exposure to LT3114 was LT3114 dose-dependent (in all three animals) and this appeared to be dose-proportional (in 2 of 3 animals). Mean C_0 was 284 ± 53.7 µg/mL and 1388 ± 127.0 µg/mL for nominal mean 10.5 mg/kg and nominal mean 51.5 mg/kg doses, respectively. Monkey #1 was observed with less exposure to LT3114 and greater elimination and clearance of LT3114 when compared to the other two monkeys in both treatments. $AUC_{0\text{-}inf}$ ranged from 62596 to 68851µg*hr/mL for two monkeys at the low dose and $AUC_{0\text{-}inf}$ ranged from 393377 to 446165 µg*hr/mL for the same two monkeys at the high dose; for monkey #1, $AUC_{0\text{-}inf}$ values were 44978 and 57641 µg*hr/mL. Terminal plasma $T_{1/2}$ ranged from 303 to 405 hours for two monkeys; for monkey #1, $T_{1/2}$ values were 89.7 and 9.58 hours following the low and high dose, respectively. Systemic clearance and apparent volume of distribution values were similar per dose level for monkey #2 and #3; Cl values ranged from 0.117 to 0.165 mL/hr/kg and V_{ss} values ranged from 61.6 to 70.3 mL/kg. For monkey #1, the Cl was about 2-7 times greater and its V_{ss} was about two times less. Mean V_c was ~37-38 mL/kg.

IX. Tables

A. Table 1 - Individual and Mean Animal Body Weights and Dosing Details

Monkey	Sex	Route	Date	Test Article	Body Weight (kg)	Syringe Weights (g)			Nominal Dose Volume		Nominal Dose Level	
						Loaded	Unloaded	Net	(mL)	(g/kg)[a]	(mg)[b]	(mg/kg)[c]
1	M	IV	27-Aug-14	LT3114	3.1	33.57	16.77	16.80	16.8	5.42	33.6	10.8
2	M	IV	27-Aug-14	LT3114	3.5	35.22	16.80	18.42	18.4	5.26	36.8	10.5
3	M	IV	27-Aug-14	LT3114	3.3	33.68	16.74	16.94	16.9	5.13	33.9	10.3
Treatment #1				Mean	3.3	34.16	16.77	17.39	17.4	5.27	34.8	10.5
				SD	0.20	0.923	0.0300	0.898	0.90	0.143	1.79	0.29
				%RSD	6.1	2.7	0.2	5.2	5.2	2.7	5.2	2.7

Monkey	Sex	Route	Date	Test Article	Body Weight (kg)	Syringe Weights (g)			Nominal Dose Volume		Nominal Dose Level	
						Loaded	Unloaded	Net	(mL)	(g/kg)[a]	(mg)[b]	(mg/kg)[c]
1	M	IV	25-Sep-14	LT3114	3.4	33.64	16.53	17.11	17.1	5.03	171	50.3
2	M	IV	25-Sep-14	LT3114	3.8	36.40	16.71	19.69	19.7	5.18	197	51.8
3	M	IV	25-Sep-14	LT3114	3.5	35.28	16.93	18.35	18.4	5.24	184	52.4
Treatment #2				Mean	3.6	35.11	16.72	18.38	18.4	5.15	184	51.5
				SD	0.21	1.388	0.200	1.290	1.290	0.108	12.9	1.08
				%RSD	5.8	4.0	1.2	7.0	7.0	2.1	7.0	2.1

[a]The nominal dose volume (g/kg) was based on the amount in grams of dosing formulation administered to the animal. This amount was calculated as the difference in weights between the pre-dose loaded dosing syringe/needle and the post-dose unloaded dosing syringe/needle.
[b]The nominal amount (mg) administered was calculated as the product of the net syringe weight and the nominal dose concentration (1.999 mg/mL for Treatment 1 and 10.001 mg/mL for Treatment 2)
[c]The nominal dose level (mg/kg) was calculated as the nominal dose concentration multiplied by the nominal dose volume.

B. Table 2 – Treatment 1 Individual and Mean LT3114 Plasma Concentration-Time Data Following a Single ~10 mg/kg Intravenous Dose of LT3114 to Male Cynomolgus Monkeys

Trt #	Monkey #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc.* (µg/mL)	Mean Conc. (µg/mL)	SD	%RSD
1	1	M	10.8	0	0.00			
1	2	M	10.5	0	0.0388			
1	3	M	10.3	0	0.00	0.0129	0.02240	173.2
1	1	M	10.8	0.5	233.5			
1	2	M	10.5	0.5	265.2			
1	3	M	10.3	0.5	321.7	273	44.7	16.3
1	1	M	10.8	3	228.5			
1	2	M	10.5	3	214.6			
1	3	M	10.3	3	239.6	228	12.5	5.5
1	1	M	10.8	6	217.6			
1	2	M	10.5	6	216.6			
1	3	M	10.3	6	220.5	218	2.0	0.9
1	1	M	10.8	12	194.6			
1	2	M	10.5	12	172.8			
1	3	M	10.3	12	195.1	188	12.7	6.8
1	1	M	10.8	24	151.0			
1	2	M	10.5	24	156.9			
1	3	M	10.3	24	163.0	157	6.0	3.8
1	1	M	10.8	48	156.5			
1	2	M	10.5	48	151.8			
1	3	M	10.3	48	127.3	145	15.7	10.8
1	1	M	10.8	72	126.9			
1	2	M	10.5	72	124.5			
1	3	M	10.3	72	102.7	118	13.3	11.3
1	1	M	10.8	120	133.5			
1	2	M	10.5	120	108.6			
1	3	M	10.3	120	100.6	114	17.2	15.0
1	1	M	10.8	180	107.8			
1	2	M	10.5	180	96.8			
1	3	M	10.3	180	100.9	102	5.6	5.5
1	1	M	10.8	240	78.4			
1	2	M	10.5	240	83.3			
1	3	M	10.3	240	75.1	78.9	4.13	5.2
1	1	M	10.8	360	60.1			
1	2	M	10.5	360	68.3			
1	3	M	10.3	360	60.0	62.8	4.76	7.6
1	1	M	10.8	480	9.10			
1	2	M	10.5	480	50.1			
1	3	M	10.3	480	48.7	36.0	23.28	64.7

*Sample assay results Below the Limit of Quantitation (BLQ=3.9 ng/mL) were reported as 0.00 µg/mL.

C. Table 3 – Treatment 2 Individual and Mean LT3114 Plasma Concentration-Time Data Following a Single ~50 mg/kg Intravenous Dose of LT3114 to Male Cynomolgus Monkeys

Trt #	Monkey #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc.* (µg/mL)	Mean Conc. (µg/mL)	SD	%RSD
2	1	M	50.3	0	0.00			
2	2	M	51.8	0	6.5			
2	3	M	52.4	0	5.9	4.13	3.592	86.9
2	1	M	50.3	0.5	1244.9			
2	2	M	51.8	0.5	1340.0			
2	3	M	52.4	0.5	1472.2	1352	114.2	8.4
2	1	M	50.3	3	1034.9			
2	2	M	51.8	3	1613.5			
2	3	M	52.4	3	1208.0	1285	297.0	23.1
2	1	M	50.3	6	1004.8			
2	2	M	51.8	6	1214.0			
2	3	M	52.4	6	1016.0	1078	117.7	10.9
2	1	M	50.3	12	778.4			
2	2	M	51.8	12	1133.6			
2	3	M	52.4	12	1143.5	1019	208.0	20.4
2	1	M	50.3	24	687.0			
2	2	M	51.8	24	1010.6			
2	3	M	52.4	24	1013.3	904	187.6	20.8
2	1	M	50.3	48	422.2			
2	2	M	51.8	48	716.2			
2	3	M	52.4	48	767.8	635	186.4	29.3
2	1	M	50.3	72	479.1			
2	2	M	51.8	72	333.1			
2	3	M	52.4	72	620.5	478	143.7	30.1
2	1	M	50.3	120	24.6			
2	2	M	51.8	120	611.9			
2	3	M	52.4	120	559.5	399	325.0	81.5
2	1	M	50.3	180	0.0423			
2	2	M	51.8	180	591.0			
2	3	M	52.4	180	534.6	375	326.1	86.9
2	1	M	50.3	240	0.00			
2	2	M	51.8	240	497.8			
2	3	M	52.4	240	500.6	333	288.2	86.6
2	1	M	50.3	360	0.00			
2	2	M	51.8	360	417.6			
2	3	M	52.4	360	409.5	276	238.8	86.6
2	1	M	50.3	480	0.00			
2	2	M	51.8	480	300.0			
2	3	M	52.4	480	321.5	207	179.7	86.8

*Sample assay results Below the Limit of Quantitation (BLQ=3.9 ng/mL) were reported as 0.00 µg/mL.

D. Table 4 - Individual and Mean Pharmacokinetic Parameters in Plasma Following a Single Intravenous Dose of LT3114 to Male Cynomolgus Monkeys

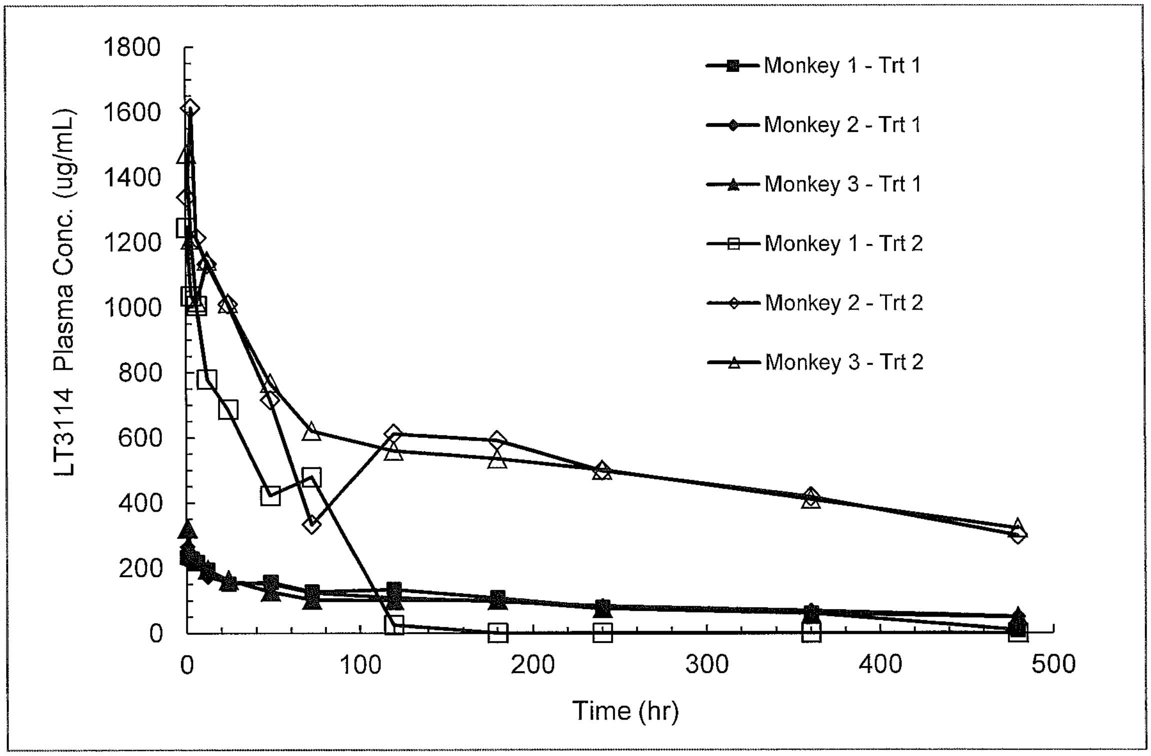
Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{max} (µg/mL)	T_{max} (hr)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
1	1	10.8	235	234	0.5	9.10	480	43301	44978	89.7	0.240	43.8	46.2
1	2	10.5	277	265	0.5	50.1	480	44908	68851	325	0.153	69.3	38.0
1	3	10.3	341	322	0.5	48.7	480	42029	62596	303	0.165	70.3	30.1
	Mean	10.5	284	273	0.5	36.0	480	43413	58808	239.2	0.186	61.1	38.1
	SD	0.29	53.7	44.7	0.00	23.28	0.0	1442.7	12379.0	130.0	0.04744	15.01	8.06
	%RSD	2.7	18.9	16.3	0.0	64.7	0.0	3.3	21.0	54.3	25.5	24.6	21.2

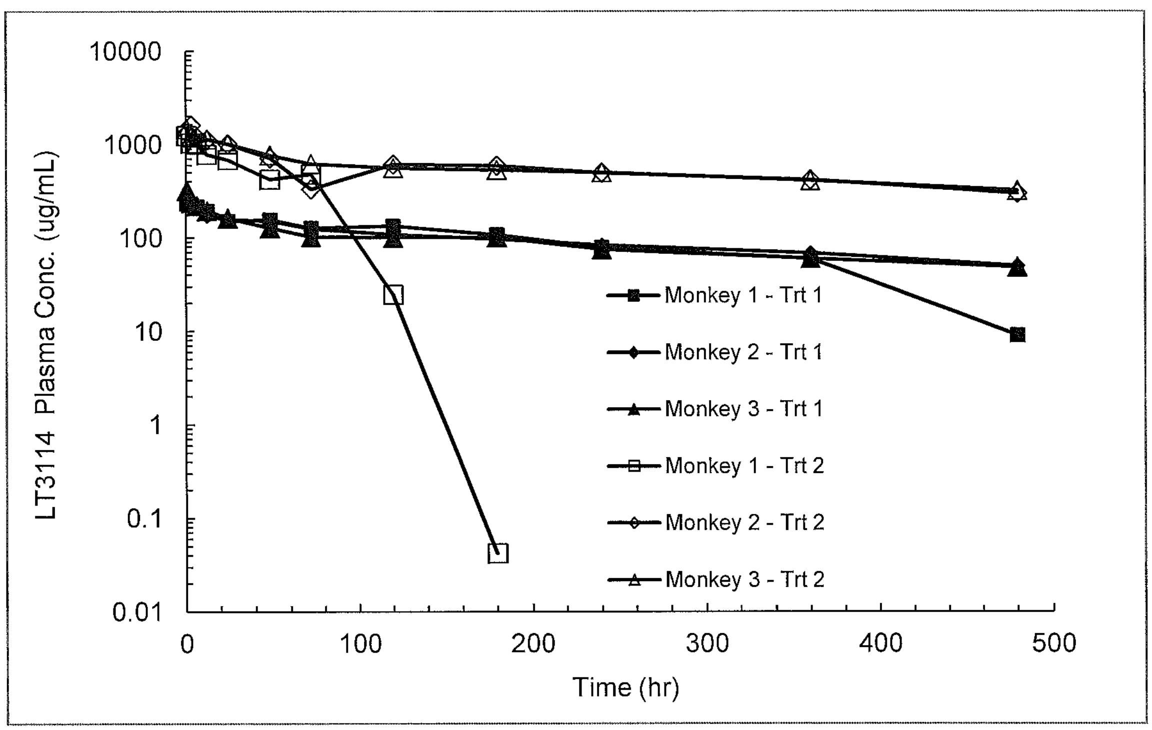
Trt #	Monkey #	Dose (mg/kg)	C_0 (µg/mL)	C_{max} (µg/mL)	T_{max} (hr)	C_{last} (µg/mL)	T_{last} (hr)	$AUC_{0\text{-}last}$ (µg*hr/mL)	$AUC_{0\text{-}inf}$ (µg*hr/mL)	$T_{1/2}$ (hr)	Cl (mL/hr/kg)	V_{ss} (mL/kg)	V_c (mL/kg)
2	1	50.3	1292	1245	0.5	0.0423	180	57640	57641	9.58	0.873	34.9	39.0
2	2	51.8	1340[a]	1614	3	300	480	251235	393377	321	0.132	61.6	38.7
2	3	52.4	1532	1472	0.5	322	480	255554	446165	405	0.117	66.6	34.2
	Mean	51.5	1388	1444	1.3	207	380	188143	299061	245	0.374	54.4	37.3
	SD	1.08	127.0	186.1	1.44	179.9	173.2	113039.5	210735.3	208.3	0.43195	17.04	2.65
	%RSD	2.1	9.1	12.9	108.3	86.7	45.6	60.1	70.5	85.0	115.5	31.3	7.1

[a] The value at dosing time (C_0) could not be determined by log-regression of the first two measurements, and was set to the first observed measurement.

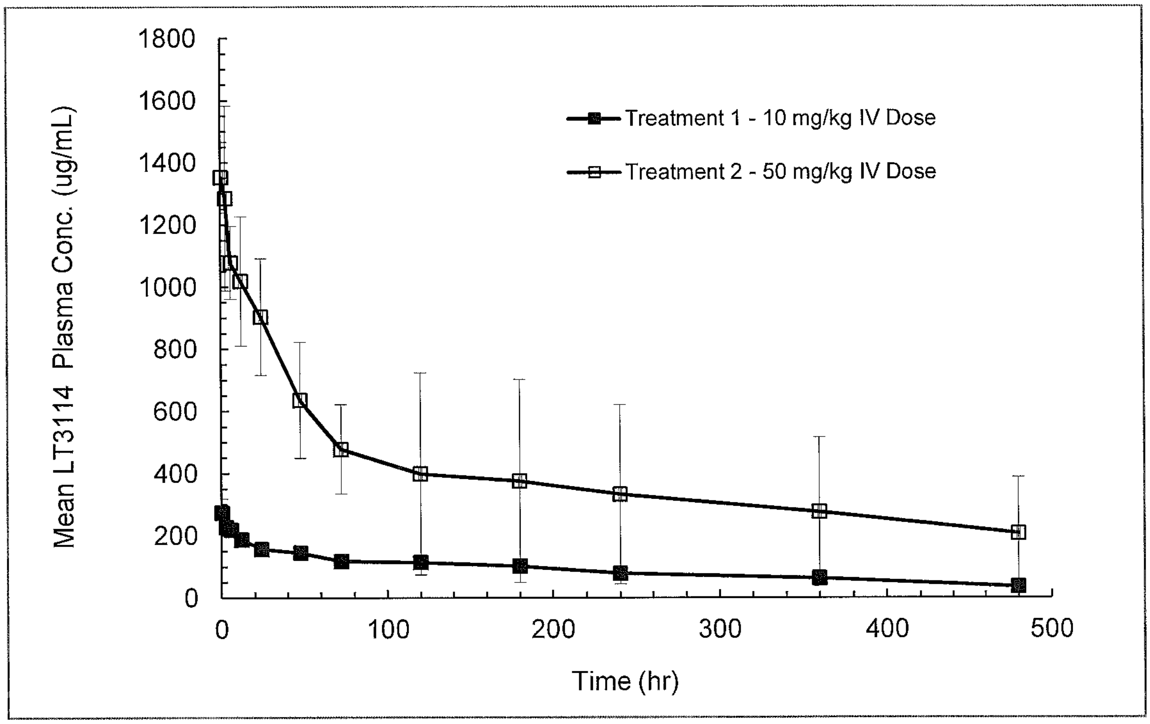
X. Figures

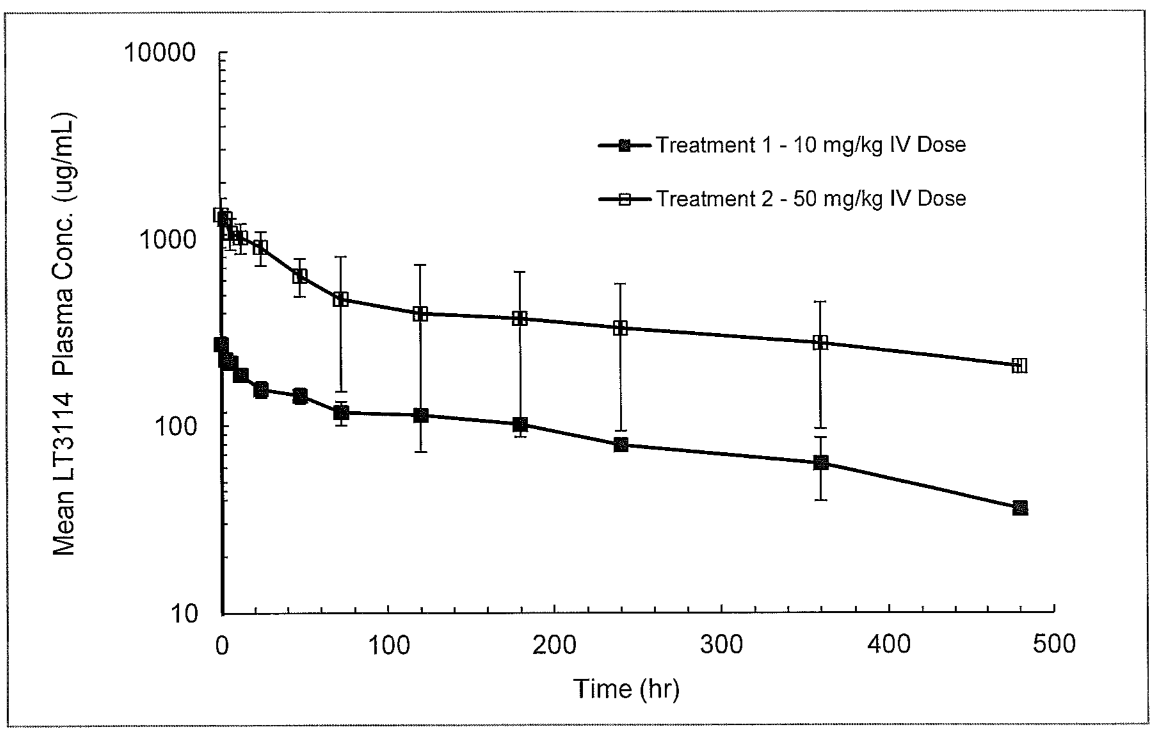
A. Figure 1 - Individual LT3114 Plasma Concentrations Plotted as a Function of Time Following a Single Intravenous Dose of LT3114 at ~10 mg/kg (Treatment 1) and ~50 mg/kg (Treatment 2) to Male Cynomolgus Monkeys (Rectilinear and Semi-Logarithmic Plots)





B. Figure 2 – Mean ± SD LT3114 Plasma Concentrations Plotted as a Function of Time Following a Single Intravenous Dose of LT3114 at ~10 mg/kg (Treatment 1) and ~50 mg/kg (Treatment 2) to Male Cynomolgus Monkeys (Rectilinear and Semi-Logarithmic Plots)





XI. Appendices

A. Appendix I—Test Article Information

Lpath Inc.	Certificate of Analysis

TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing	REVISION NO. 1.0.0

Name	Part Number		Lot Number
Lpathomab-DS, 30 mg/mL	**TBD**		GA121459
Formulation Buffer			**Storage Temperature**
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0			2 to 8°C
Production Date	**Test Date**	**Retest Date**	**Intended Use**
JUN 2014	JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer	**Manufacturer Part Number**		**Manufacturer Lot Number**
Gallus-MO, St. Louis, MO	TBD		**Gallus-MO 103948**

Attribute	Test	Specification	Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2	5.1
	CGE (Non-reduced)	Major band at 150 kDa	Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	iCE	Comparable to RS	Conforms
		Minor Peak at pI ~8.3 Major Peak at pI ~8.4 Minor Peak at pI ~8.6 Faint Peak at pI ~8.7	15% 60% 13% 4%
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL	32.1 mg/mL
	LPA Binding	100% ±40% of RS	89%

Lpath Inc.	Certificate of Analysis

TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing	REVISION NO. 1.0.0

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL[1]
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance: ______________________ **Date: 22AUG2014**

Acting Head of QA - Frieder K. Hofmann, PhD

[1] At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

B. Appendix II—Protocol, Protocol Amendment and Protocol Deviations

CALVERT LABS
Customized. Responsive. **Proven.**

Study Protocol

Title:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)
Calvert Study No.:	0875SL35.001
Testing Facility:	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447
Study Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121

I. Table Of Contents

II. Introduction

A. Title

A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)

B. Objective

To evaluate the pharmacokinetics of LT3114 in male non-naïve Cynomolgus monkeys after a single intravenous dose of LT3114 at two different dose levels.

C. Regulatory Compliance

This is a non-regulated study. However, it will be run according to the Standard Operating Procedures (SOPs) of Calvert. There will be no formal involvement of Calvert Quality Assurance.

D. Calvert Study Number

0875SL35.001

E. Testing Facility

Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

F. Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

G. Study Director

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Phone: (570) 585-2222

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H. Study Monitor

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14 Joseph Court
Monmouth Junction, NJ 08852
Phone: (732) 274-9562
Mobile: (732) 822-6101
E-mail: mwmconsultinggroup@gmail.com

I. Sponsor Representative

Gary Woodnutt, Ph.D.
Senior Vice President Research
Lpath Inc
4025 Sorrento Valley Blvd
San Diego, CA 92121

J. Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis

Victor J. Johnson, Ph.D.
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Email: vjohnson@brt-labs.com

K. Key Study Dates

Proposed First Day of Dosing:	27 Aug 2014
Proposed Experimental Completion Date (In-Life Portion):	21 Oct 2014

III. Materials and Methods

A. Test Article and Test Article Formulations

1. Test Article

Identification:	LT3114
Lot/Batch No.:	103948
Physical Description:	Clear colorless liquid
Storage Conditions:	Stored refrigerated (2-8°C) and protected from light.

2. Dose Preparations

Before dose preparation, 2 ml of a 10% polysorbate 80 solution (provided by the Sponsor; Batch# 104012-001, MM#306191), will be added aseptically to each liter of formulation buffer (also supplied by the Sponsor and in sterile containers; Batch# 104011-001, MM#306190), and mixed gently to ensure distribution throughout the volume of buffer. This diluent will then be used for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations. The Sponsor will provide the test article already formulated at 32.1 mg/ml in sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared. Do not shake the formulation. Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

3. Dose Formulation Samples

Triplicate 1-ml samples of each dosing solution will be obtained. These samples will be stored at refrigerated temperatures (2-8°C) until shipment to the Sponsor-designee.

4. Formulated Test Article Analysis

Dose formulation samples will be shipped to and analytical evaluation will be done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. Concentration of LT3114 will be measured.

5. Accountability and Disposition

Unused test article will be returned to the Sponsor or designee at the completion of this study or, if necessary, retained for use on related future studies. The Sponsor will be notified in advance of shipping and a transmittal letter will accompany the shipment. The material will be packed in a suitable container to maintain the conditions specified by the Sponsor during transit plus an adequate margin of safety to account for any possible transit delays. If material is returned, the Sponsor will provide shipping instructions.

B. Test System (Animals and Animal Care)

1. *Description*

Species: Monkey

Breed: Cynomolgus (Macaca fascicularis)

Total Number: 3

Gender: Male

Age Range: ~4 to 4.5 years at start of dosing; records of dates of birth for animals used in this study will be retained in the Calvert archives.

Body Weight Range: 3.2 to 4.2 kg

Animal Source: SNBL (Texas); Cambodia origin

Experimental History: Purpose-bred and experimentally non-naïve at the outset of the study but naïve to monoclonal antibody administration.

Identification: Chest tattoo and cage card

2. *Rationale for Choice of Species and Number of Animals*

The cynomolgus monkey is a standard non-rodent species used in toxicology studies based upon the substantial amounts of published historical data (1). The test article is a humanized antibody. Nonclinical studies designed to test humanized antibodies are generally conducted in nonhuman primates rather than other large animal species because the antibodies are likely to be reactive in nonhuman primate (2). No previous research has been conducted with the test article in nonhuman primates. The same three animals will be used per dose level in crossover fashion to meet the minimum numbers for descriptive statistical analysis of the data.

3. *Husbandry*

Housing:	Animals were housed during acclimation in compliance with USDA Guidelines (7 CFR 2.22, 2.80 and 371.2) and Calvert SOP VET-44. Animals will be housed accordingly during the in-life portion of the study. The room in which the animals will be kept will be documented in the study records. No other species will be kept in the same room.
Lighting:	12 hours light/12 hours dark except when room lights will be turned on during the dark cycle due to in-life procedures.
Room Temperature:	18 to 29°C
Relative Humidity:	30 to 70%
Food:	All animals will have access to Harlan Teklad Primate Diet (certified) or equivalent approximately 4% of their body weight daily. Chemical analysis of the feed will be performed by the manufacturer. Results of these analyses will be kept on file at Calvert. Certified Primatreats®, fruit and food suitable for human consumption are also given as a supplement at least once daily. The lot number(s) and specifications of each lot of Primatreats® and Certified Primate Diet (Teklad) used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet will be limited to that performed by the manufacturer, records of which will be maintained in the Calvert archives.
Water:	Water will be available *ad libitum* to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the

results of this study. Results of the water analysis will be maintained in the Calvert archives.

Acclimation: Study animals have been acclimated to their housing for a minimum of 5 months.

4. *Prestudy Health Screen and Selection Criteria*

All animals received for this study have been assessed as to their general health by a member of the veterinary staff or other authorized personnel. During the acclimation period, each animal was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that are determined to be suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

5. *Assignment to Study Groups*

Because of the small number of animals in this study, animals will be manually assigned from an appropriate pool of animals.

6. *Humane Care of Animals*

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge, this study is not unnecessary or duplicative.

C. Dose Administration

1. Treatments and Dosing Details

Trt	# of Animals and Sex	Route of Administration	Dose Conc. (mg/ml)	Dose Volume (ml/kg)	Dose Level (mg/kg)
1	3 males[a]	IV	2	5	10
2	3 males[a]	IV	6	5	30

[a] The same three animals will be used in each treatment; there will be at least a 4-week washout/recovery period between the administration of each single bolus intravenous treatment. Treatment 1 will be administered on Day 1 followed by treatment 2 to be administered at least 4 weeks (28 days) later.

2. Dosing

Route: Intravenous

Frequency: Single bolus intravenous administration per treatment.

Procedure: A single intravenous dose will be administered into a saphenous (left or right) vein per treatment. Each intravenous dose will be administered as a manual slow bolus injection over a period of approximately ~3 minutes.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans. Dose levels were selected by the Sponsor. The anticipated lowest pharmacologically active dose is 30 mg/kg.

D. In-Life Observations and Measurements

1. Clinical Observations

Frequency: Animals will be observed at least once daily and at all times of blood sampling. Any abnormal clinical signs will be recorded.

2. Body Weight

Frequency: Animals will be weighed on each day of test article administration prior to dosing.

E. Pharmacokinetics

1. Dosing Syringes

The actual dose administered to the animal will be the difference between the loaded and unloaded dosing syringe weights. Syringe weights will be measured and recorded to at least two decimal places.

2. Blood Sampling

Following each dose administration, serial blood samples (~1 ml each) will be collected by venipuncture of a femoral or saphenous vein (excluding the site used for intravenous dosing) at 0, 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours. Blood samples will be collected into chilled tubes containing K_2EDTA as the anti-coagulant. Blood tubes will be placed on chill packs until being centrifuged within 30 minutes of collection.

3. Plasma Harvesting

Plasma will be harvested following centrifugation of the chilled blood samples at 3000 RPM for 10 minutes at +4 °C. Derived plasma will be transferred into prelabeled plastic vials (label designating Calvert study number, animal number, route, dose level/test article, date of collection, time of collection, and sample type) and stored frozen at approximately -80 °C.

4. Sample Shipment

Frozen plasma samples will be shipped frozen on dry ice via courier overnight delivery to and bioanalytical evaluation will be done under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, 120 First Flight Lane, Morrisville, NC 27560. A cover letter, sample transmittal form, and sample log will be sent with the samples.

F. Terminal Procedures

1. Termination

a) Scheduled Sacrifice

No terminal sacrifice is planned in this study. Animals will be transferred back into the primate colony at Calvert following collection of the final blood sample of the final treatment in this study. However, moribund animals may be euthanized by intravenous barbiturate overdose for humane reasons following consultation with a veterinarian, and, ***if possible***, with the Sponsor.

2. *Gross Necropsy*

Any animal found dead or euthanized in a moribund condition will be subjected to a gross pathological exam at additional expense to the Sponsor. Any observations will be noted and any gross lesions will be preserved in Formalin for potential future analysis. Following necropsy and collection of any gross lesions the carcass will be discarded.

IV. Records and Reports

A. Data Analysis

Data analysis will consist of descriptive statistical analysis of the animal body weights and dosing details per treatment. Animal observations and blood sampling times (actual and elapsed) will be recorded in the raw data. Bioanalysis of the plasma samples will be the responsibility of Burleson Research Technologies (Morrisville, NC 27560). Pharmacokinetic analysis of the plasma sample assay results will be the responsibility of Calvert. Using noncompartmental analysis and WinNonlin Version 5.3 software, individual pharmacokinetics parameters will include $AUC_{0\text{-}inf}$, $AUC_{0\text{-}last}$, C_{max}, Cl, C_{last}, C_0, T_{last}, T_{max}, $T_{1/2}$, V_{ss}, and V_c. Note: If the terminal half-life cannot be reliably calculated, the dependent PK parameters of $AUC_{0\text{-}inf}$, Cl, V_{ss}, and V_c will not be reported.

B. Storage of Records

Test article tracking, in-life data, protocol, protocol amendments (if applicable), technical notes (if applicable), and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. Within 90 days following the submission of the draft report, if there are no client comments generated by the Sponsor and/or Study Monitor, the Sponsor/Study Monitor will be notified and the report may be finalized and archived according to the terms stated in the protocol. After 2 years following report finalization, the Sponsor will be contacted to determine final disposition of all study materials.

V. Miscellaneous

A. Confidentiality Statement

The information contained herein is for the personal use of the intended recipient(s).

B. References

1. Animal Models in Toxicology. Second Edition. Edited by Shayne C. Gad. Chapter 9 – Primates. *Metabolism*. Pages 716-723.

2. Chapman K., Pullen N., Cooney L., Dempster M., et. al. (2009). Preclinical Development of Monoclonal Antibodies. mAbs, I: 505-516.

VI. Protocol Approval Signatures

William G. Tuman 29 Jul 2014

Study Director's Signature Date

William G. Tuman, M.S.

Calvert Laboratories, Inc.

22 JUL 2014

Scientific Management Date

Calvert Laboratories, Inc.

Brian Klatt 21 Jul 2014

IACUC Date

Calvert Laboratories, Inc.

Marlene Modi, Ph.D. Date

Study Monitor

Gary Woodnutt, PhD Date

Sponsor Representative

VI. Protocol Approval Signatures

Study Director's Signature Date
William G. Tuman, M.S.
Calvert Laboratories, Inc.

Scientific Management Date
Calvert Laboratories, Inc.

IACUC Date
Calvert Laboratories, Inc.

Marlene Modi 28 July 2014
Marlene Modi, Ph.D. Date
Study Monitor

28th July 2014
Gary Woodnutt, PhD Date
Sponsor Representative

CALVERT LABS

Customized. Responsive. **Proven.**

PROTOCOL AMENDMENT

STUDY NUMBER: 0875SL35.001

STUDY TITLE: A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)

AMENDMENT NUMBER: 1

EFFECTIVE DATE: 4 Aug 2014

PROTOCOL SECTION CHANGES: ***Page 10***

III. Materials and Methods

C. Dose Administration

1. Treatments and Dosing Details

Trt	# of Animals and Sex	Route of Administration	Dose Conc. (mg/ml)	Dose Volume (ml/kg)	Dose Level (mg/kg)
1	3 males[a]	IV	2	5	10
2	3 males[a]	IV	~~6~~ **10**	5	~~30~~ **50**

[a] The same three animals will be used in each treatment; there will be at least a 4-week washout/recovery period between the administration of each single bolus intravenous treatment. Treatment 1 will be administered on Day 1 followed by treatment 2 to be administered at least 4 weeks (28 days) later.

3. Justification for Route and Dose Levels

The intravenous route was chosen as it is the intended route of administration in humans. Dose levels were selected by the Sponsor **based upon the findings of a dose range finding study previously conducted by the Sponsor**. The anticipated lowest pharmacologically active dose is 30 mg/kg.

REASON FOR CHANGES:

The Sponsor requested the dose level for Treatment 2 to be changed from 30 mg/kg to 50 mg/kg based upon the findings of a dose range finding study previously conducted by the Sponsor.

SIGNATURES:

William G. Tuman — 4 Aug 2014

William G. Tuman, M.S.
Study Director
Calvert Laboratories, Inc.

Date

Steven D. Abner... — 4 Aug 2014

Scientific Management
Calvert Laboratories, Inc.

Date

Brian Klot — 4 Aug 2014

IACUC
Calvert Laboratories, Inc.

Date

Marlene Modi, Ph.D.
Study Monitor

Date

Gary Woodnutt, Ph.D.
Study Representative

Date

REASON FOR CHANGES:

The Sponsor requested the dose level for Treatment 2 to be changed from 30 mg/kg to 50 mg/kg based upon the findings of a dose range finding study previously conducted by the Sponsor.

SIGNATURES:

William G. Tuman, M.S.
Study Director
Calvert Laboratories, Inc.

Date

Scientific Management
Calvert Laboratories, Inc.

Date

IACUC
Calvert Laboratories, Inc.

Date

Marlene Modi

Marlene Modi, Ph.D.
Study Monitor

01 August 2014
Date

Gary Woodnutt

Gary Woodnutt, Ph.D.
Study Representative

1st August 2014
Date

Protocol Deviations

1. The protocol-specified animal body weight range was 3.2 to 4.2 kg. One animal was below this weight range (3.1 kg) prior to dose administration for Treatment 1. This slight deviation in the body weight range was not considered to have an impact on the outcome or integrity of the study since the animals were within the specified age range and the dose administered was normalized for body weight.

2. The protocol specified that the monkeys used on this study would be non-naïve. Due to a postponement/scheduling change for a separate study that was to use these same colony animals, the monkeys used on this study were subsequently naïve. This deviation did not impact the integrity or outcome of the study.

C. Appendix III—Dosing and Blood Sampling Times (Actual and Elapsed) and Clinical Observations

0875SL35.001

Monkey #	Trt #	Sex	Date	Nom. Dose (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1	1	M	27-Aug-14	10	IV	7:26:36	7:29:36	0	7:09:47		Normal	7:15	7:32
1	1	M	27-Aug-14	10	IV			0.5	7:56:03	0:29:27	Normal	8:08	8:20
1	1	M	27-Aug-14	10	IV			3	10:26:30	2:59:54	Normal	10:39	10:59
1	1	M	27-Aug-14	10	IV			6	13:27:01	6:00:25	Normal	13:37	13:50
1	1	M	27-Aug-14	10	IV			12	19:26:28	11:59:52	Normal	19:36	19:50
1	1	M	28-Aug-14	10	IV			24	7:26:52	24:00:16	Normal	7:37	7:54
1	1	M	29-Aug-14	10	IV			48	7:26:41	48:00:05	Normal	7:38	7:50
1	1	M	30-Aug-14	10	IV			72	7:26:10	71:59:34	Normal	7:39	8:07
1	1	M	1-Sep-14	10	IV			120	7:28:16	120:01:40	Normal	7:38	7:52
1	1	M	3-Sep-14	10	IV			180	19:26:00	179:59:24	Normal	19:36	19:48
1	1	M	6-Sep-14	10	IV			240	7:26:21	239:59:45	Normal	7:36	7:49
1	1	M	11-Sep-14	10	IV			360	7:26:58	360:00:22	Normal	7:37	7:49
1	1	M	16-Sep-14	10	IV			480	7:26:46	480:00:10	Normal	7:40	7:57
2	1	M	27-Aug-14	10	IV	7:31:53	7:34:53	0	7:08:31		Normal	7:15	7:32
2	1	M	27-Aug-14	10	IV			0.5	8:01:21	0:29:28	Normal	8:08	8:20
2	1	M	27-Aug-14	10	IV			3	10:31:29	2:59:36	Normal	10:39	10:59
2	1	M	27-Aug-14	10	IV			6	13:31:55	6:00:02	Normal	13:37	13:50
2	1	M	27-Aug-14	10	IV			12	19:30:21	11:58:28	Normal	19:36	19:50
2	1	M	28-Aug-14	10	IV			24	7:31:31	23:59:38	Normal	7:37	7:54
2	1	M	29-Aug-14	10	IV			48	7:31:38	47:59:45	Normal	7:38	7:50
2	1	M	30-Aug-14	10	IV			72	7:30:37	71:58:44	Normal	7:39	8:07
2	1	M	1-Sep-14	10	IV			120	7:31:23	119:59:30	Normal	7:38	7:52
2	1	M	3-Sep-14	10	IV			180	19:31:02	179:59:09	Normal	19:36	19:48
2	1	M	6-Sep-14	10	IV			240	7:31:16	239:59:23	Normal	7:36	7:49
2	1	M	11-Sep-14	10	IV			360	7:31:28	359:59:35	Normal	7:37	7:49
2	1	M	16-Sep-14	10	IV			480	7:31:50	479:59:57	Normal	7:40	7:57

Monkey #	Trt #	Sex	Date	Nom. Dose (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
3	1	M	27-Aug-14	10	IV	7:35:52	7:38:52	0	7:12:50		Normal	7:15	7:32
3	1	M	27-Aug-14	10	IV			0.5	8:05:40	0:29:48	Normal	8:08	8:20
3	1	M	27-Aug-14	10	IV			3	10:35:51	2:59:59	Normal	10:39	10:59
3	1	M	27-Aug-14	10	IV			6	13:35:31	5:59:39	Normal	13:37	13:50
3	1	M	27-Aug-14	10	IV			12	19:35:52	12:00:00	Normal	19:36	19:50
3	1	M	28-Aug-14	10	IV			24	7:35:36	23:59:44	Normal	7:37	7:54
3	1	M	29-Aug-14	10	IV			48	7:35:58	48:00:06	Normal	7:38	7:50
3	1	M	30-Aug-14	10	IV			72	7:35:27	71:59:35	Normal	7:39	8:07
3	1	M	1-Sep-14	10	IV			120	7:35:01	119:59:09	Normal	7:38	7:52
3	1	M	3-Sep-14	10	IV			180	19:35:05	179:59:13	Normal	19:36	19:48
3	1	M	6-Sep-14	10	IV			240	7:35:02	239:59:10	Normal	7:36	7:49
3	1	M	11-Sep-14	10	IV			360	7:35:47	359:59:55	Normal	7:37	7:49
3	1	M	16-Sep-14	10	IV			480	7:35:59	480:00:07	Normal	7:40	7:57

Monkey #	Trt #	Sex	Date	Nom. Dose (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
1	2	M	25-Sep-14	50	IV	6:59:03	7:02:03	0	6:36:31		Normal	6:45	7:16
1	2	M	25-Sep-14	50	IV			0.5	7:29:20	0:30:17	Normal	7:41	7:51
1	2	M	25-Sep-14	50	IV			3	9:59:11	3:00:08	Normal	10:12	10:27
1	2	M	25-Sep-14	50	IV			6	12:59:04	6:00:01	Normal	13:12	13:41
1	2	M	25-Sep-14	50	IV			12	18:59:09	12:00:06	Normal	19:10	19:22
1	2	M	26-Sep-14	50	IV			24	6:59:19	24:00:16	Normal	7:10	7:22
1	2	M	27-Sep-14	50	IV			48	6:59:09	48:00:06	Normal	7:13	7:26
1	2	M	28-Sep-14	50	IV			72	6:59:04	72:00:01	Normal	7:12	8:49
1	2	M	30-Sep-14	50	IV			120	6:59:48	120:00:45	Normal	7:11	7:23
1	2	M	2-Oct-14	50	IV			180	18:59:02	179:59:59	Normal	19:11	19:23
1	2	M	5-Oct-14	50	IV			240	7:00:13	240:01:10	Normal	7:12	8:06
1	2	M	10-Oct-14	50	IV			360	6:59:09	360:00:06	Normal	7:15	7:27
1	2	M	15-Oct-14	50	IV			480	6:59:23	480:00:20	Normal	7:11	7:25
2	2	M	25-Sep-14	50	IV	7:04:41	7:07:41	0	6:39:11		Normal	6:45	7:16
2	2	M	25-Sep-14	50	IV			0.5	7:34:27	0:29:46	Normal	7:41	7:51
2	2	M	25-Sep-14	50	IV			3	10:04:35	2:59:54	Normal	10:12	10:27
2	2	M	25-Sep-14	50	IV			6	13:03:53	5:59:12	Normal	13:12	13:41
2	2	M	25-Sep-14	50	IV			12	19:03:39	11:58:58	Normal	19:10	19:22
2	2	M	26-Sep-14	50	IV			24	7:04:00	23:59:19	Normal	7:10	7:22
2	2	M	27-Sep-14	50	IV			48	7:04:13	47:59:32	Normal	7:13	7:26
2	2	M	28-Sep-14	50	IV			72	7:03:48	71:59:07	Normal	7:12	8:49
2	2	M	30-Sep-14	50	IV			120	7:05:06	120:00:25	Normal	7:11	7:23
2	2	M	2-Oct-14	50	IV			180	19:04:36	179:59:55	Normal	19:11	19:23
2	2	M	5-Oct-14	50	IV			240	7:04:32	239:59:51	Normal	7:12	8:06
2	2	M	10-Oct-14	50	IV			360	7:04:13	359:59:32	Normal	7:15	7:27
2	2	M	15-Oct-14	50	IV			480	7:04:10	479:59:29	Normal	7:11	7:25

Monkey #	Trt #	Sex	Date	Nom. Dose (mg/kg)	Route	Dose Time		Nominal Time (hr)	Actual Time	Elapsed Time (hr:min:sec)	Clinical Observations	Time Placed In	
						Start	End					Centrifuge	Freezer
3	2	M	25-Sep-14	50	IV	7:10:54	7:13:54	0	6:41:35		Normal	6:45	7:16
3	2	M	25-Sep-14	50	IV			0.5	7:40:00	0:29:06	Normal	7:41	7:51
3	2	M	25-Sep-14	50	IV			3	10:10:25	2:59:31	Normal	10:12	10:27
3	2	M	25-Sep-14	50	IV			6	13:09:59	5:59:05	Normal	13:12	13:41
3	2	M	25-Sep-14	50	IV			12	19:10:02	11:59:08	Normal	19:10	19:22
3	2	M	26-Sep-14	50	IV			24	7:10:09	23:59:15	Normal	7:10	7:22
3	2	M	27-Sep-14	50	IV			48	7:10:00	47:59:06	Normal	7:13	7:26
3	2	M	28-Sep-14	50	IV			72	7:10:15	71:59:21	Normal	7:12	8:49
3	2	M	30-Sep-14	50	IV			120	7:10:03	119:59:09	Normal	7:11	7:23
3	2	M	2-Oct-14	50	IV			180	19:10:58	180:00:04	Normal	19:11	19:23
3	2	M	5-Oct-14	50	IV			240	7:10:12	239:59:18	Normal	7:12	8:06
3	2	M	10-Oct-14	50	IV			360	7:10:50	359:59:56	Normal	7:15	7:27
3	2	M	15-Oct-14	50	IV			480	7:10:00	479:59:06	Normal	7:11	7:25

D. Appendix IV—Individual LT3114 Monkey Plasma Concentration-Time Data

Trt #	Monkey #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (ug/mL)
1	1	M	10.8	0	0.00
1	1	M	10.8	0.5	233.5
1	1	M	10.8	3	228.5
1	1	M	10.8	6	217.6
1	1	M	10.8	12	194.6
1	1	M	10.8	24	151.0
1	1	M	10.8	48	156.5
1	1	M	10.8	72	126.9
1	1	M	10.8	120	133.5
1	1	M	10.8	180	107.8
1	1	M	10.8	240	78.4
1	1	M	10.8	360	60.1
1	1	M	10.8	480	9.10
1	2	M	10.5	0	0.0388
1	2	M	10.5	0.5	265.2
1	2	M	10.5	3	214.6
1	2	M	10.5	6	216.6
1	2	M	10.5	12	172.8
1	2	M	10.5	24	156.9
1	2	M	10.5	48	151.8
1	2	M	10.5	72	124.5
1	2	M	10.5	120	108.6
1	2	M	10.5	180	96.8
1	2	M	10.5	240	83.3
1	2	M	10.5	360	68.3
1	2	M	10.5	480	50.1
1	3	M	10.3	0	0.00
1	3	M	10.3	0.5	321.7
1	3	M	10.3	3	239.6
1	3	M	10.3	6	220.5
1	3	M	10.3	12	195.1
1	3	M	10.3	24	163.0
1	3	M	10.3	48	127.3
1	3	M	10.3	72	102.7
1	3	M	10.3	120	100.6
1	3	M	10.3	180	100.9
1	3	M	10.3	240	75.1
1	3	M	10.3	360	60.0
1	3	M	10.3	480	48.7

Sample assay results Below the Limit of Quantitation (BLQ=3.9 ng/mL) were reported as 0.00 μg/mL.

Trt #	Monkey #	Sex	Dose (mg/kg)	Time (hr)	LT3114 Conc. (ug/mL)
2	1	M	50.3	0	0.00
2	1	M	50.3	0.5	1244.9
2	1	M	50.3	3	1034.9
2	1	M	50.3	6	1004.8
2	1	M	50.3	12	778.4
2	1	M	50.3	24	687.0
2	1	M	50.3	48	422.2
2	1	M	50.3	72	479.1
2	1	M	50.3	120	24.6
2	1	M	50.3	180	0.0423
2	1	M	50.3	240	0.00
2	1	M	50.3	360	0.00
2	1	M	50.3	480	0.00
2	2	M	51.8	0	6.5
2	2	M	51.8	0.5	1340.0
2	2	M	51.8	3	1613.5
2	2	M	51.8	6	1214.0
2	2	M	51.8	12	1133.6
2	2	M	51.8	24	1010.6
2	2	M	51.8	48	716.2
2	2	M	51.8	72	333.1
2	2	M	51.8	120	611.9
2	2	M	51.8	180	591.0
2	2	M	51.8	240	497.8
2	2	M	51.8	360	417.6
2	2	M	51.8	480	300.0
2	3	M	52.4	0	5.9
2	3	M	52.4	0.5	1472.2
2	3	M	52.4	3	1208.0
2	3	M	52.4	6	1016.0
2	3	M	52.4	12	1143.5
2	3	M	52.4	24	1013.3
2	3	M	52.4	48	767.8
2	3	M	52.4	72	620.5
2	3	M	52.4	120	559.5
2	3	M	52.4	180	534.6
2	3	M	52.4	240	500.6
2	3	M	52.4	360	409.5
2	3	M	52.4	480	321.5

Sample assay results Below the Limit of Quantitation (BLQ=3.9 ng/mL) were reported as 0.00 µg/mL.

E. Appendix V—Analytical/Bioanalytical Subreport

STUDY PHASE REPORT

STUDY TITLE:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)
REPORT TITLE:	Determination of LT3114 in Dose Formulation Samples and Monkey Plasma Pharmacokinetic Samples
BRT Reference No.:	BRT 20140726
Author:	Victor J. Johnson, PhD (BRT)
Dose Formulation and PK Analysis Laboratory:	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Testing Facility:	Calvert Laboratories, Inc. Scott Technology Park 130 Discovery Drive Scott Township, PA 18447
Study Number:	0875SL35.001
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121

Report Date: 04 December 2014

COMPLIANCE – NON-GLP STUDY

Although this study was performed as indicated in the study protocol and applicable BRT standard operating procedures (SOPs), it was investigational in nature and was not expected to conform to good laboratory practice (GLP) standards.

APPROVALS

Victor J. Johnson, PhD Principal Investigator Burleson Research Technologies, Inc.	04 Dec 14 Date
Gary R. Burleson, PhD President (Management) Burleson Research Technologies, Inc.	04 Dec 2014 Date

Report Review:

Florence G. Burleson, PhD Director of Laboratory Operations Burleson Research Technologies, Inc.	04 Dec 2014 Date

TABLE OF CONTENTS

STUDY INFORMATION

Determination of LT3114 in Dose Formulation Samples and Monkey Plasma Pharmacokinetic Samples

Study Title:	A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)
Testing Facility:	Calvert Laboratories, Inc.
Testing Facility Study No:	0875SL35.001
Study Director:	William G. Tuman, M.S. (Calvert)
Sponsor:	Lpath Incorporated 4025 Sorrento Valley Blvd. San Diego, CA, 92121
Test Site (Dose Formulation Analysis and Plasma PK Analysis):	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Test Site Reference No.:	BRT 20140726
Principal Investigator:	Victor J. Johnson, PhD (BRT)
Analysts:	Kristen Fisher (BRT) Kim Barfield (BRT)
Study Initiation Date:	29 Jul 14
Sample Receipt Date:	17 Sep 14 (dose formulation and TK samples for low dose) 16 Oct 14 (dose formulation and TK samples for high dose)
Experimental Start Date:	18 Sep 14
Experimental Completion Date:	22 Oct 14
Study Phase Completion Date:	04 Dec 14

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples and monkey plasma pharmacokinetic (PK) samples generated by Calvert Study No. 0875SL35.001: "A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)".

Dose formulation samples were provided for two dose concentrations, 2 mg/mL and 10 mg/mL. Dose formulation samples were collected from each dosing solution in triplicate. The mean LT3114 concentration for the 2 mg/mL dose formulation was 2.2 mg/mL (2.05 – 2.31 mg/mL). The mean LT3114 concentration for the 10 mg/mL dose formulation was 10.2 mg/mL (9.7 – 10.7 mg/mL). The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision.

Monkeys were administered Treatment #1 (10 mg/kg) followed by a washout period of approximately 4 weeks and then administered Treatment #2 (50 mg/kg). LT3114 was not detectable in the plasma PK samples collection from monkeys 1 and 3 prior to administration of Treatment #1 (10 mg/kg). The sample collected from monkey 2 prior to administration of Treatment #1 showed an apparent concentration of 0.039 µg/mL of LT3114 which is consistent with minimal matrix interference as observed at the MRD (1:10 dilution) during method transfer (BRT 20140420). LT3114 was detectable in all post-dose plasma PK samples collected following administration of Treatment #1 and concentrations ranged from 9.1 µg/mL to 321.7 µg/mL.

LT3114 was detectable in the Treatment #2 (50 mg/kg) pre-dose plasma PK samples from monkeys 2 (6.5 µg/mL) and 3 (5.9 µg/mL), but was not detectable in the pre-dose plasma PK sample from monkey 1. LT3114 was detectable in all post-dose plasma PK samples from monkeys 2 and 3 throughout the PK time course. However, LT3114 concentration in the plasma of monkey 1 dropped to 0.042 µg/mL at 180 hours post-dose and was undetectable in samples obtained at 240, 360, and 480 hours post-dose. LT3114 concentrations in the plasma PK samples following Treatment #2 ranged from 5.9 µg/mL to 1613.5 µg/mL.

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples and monkey plasma pharmacokinetic (PK) samples in support of the study entitled "A Pharmacokinetic Study of LT3114 Administered Intravenously in Cynomolgus Monkeys (non-GLP)".

STUDY DESIGN

Dose Formulation Samples

Dose formulations (2, and 10 mg/mL) were prepared by Calvert on 27 Aug 14, and 24 Sep 14 according to the study protocol using LT3114 in a citrate buffer formulation (32.1 mg/mL) and citrate formulation buffer supplied by the Sponsor. Triplicate 1 mL samples were collected on 27 Aug 14 (Treatment #1, 2 mg/mL dose level), and 24 Sep 14 (Treatment #2, 10 mg/mL dose level) from each dosing solution and were stored at 2-8°C until single aliquots from each sample were shipped, chilled on ice packs, to BRT (Table 1). Dose formulation samples were logged into BRT inventory and stored at 2-8°C until analyzed for LT3114 concentration. Samples were returned to 2-8°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received at BRT.

Date of Collection	Date of Receipt at BRT	Aliquots	Description
27 Aug 14	17 Sep 14	1	Dose Formulation for LT3114, 2 mg/mL (1/3)
27 Aug 14	17 Sep 14	1	Dose Formulation for LT3114, 2 mg/mL (2/3)
27 Aug 14	17 Sep 14	1	Dose Formulation for LT3114, 2 mg/mL (3/3)
24 Sep 14	16 Oct 14	1	Dose Formulation for LT3114, 10 mg/mL (1/3)
24 Sep 14	16 Oct 14	1	Dose Formulation for LT3114, 10 mg/mL (2/3)
24 Sep 14	16 Oct 14	1	Dose Formulation for LT3114, 10 mg/mL (3/3)

Monkey Plasma Pharmacokinetic Samples

For the assessment of LT3114 pharmacokinetics, 3 male Cynomolgus monkeys were administered LT3114 in a citrate buffer formulation at 10 mg/kg (Treatment #1) or 50 mg/kg (Treatment #2). Route of administration was intravenous injection as a manual slow bolus injection over a period of approximately 3 minutes via the saphenous vein. Monkeys were administered Treatment #1 (10 mg/kg) on 27 Aug 14 and whole blood samples were obtained by saphenous vein puncture from all monkeys at pre-dose (0 hour), 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose (HrPD). Blood was collected into chilled K_2EDTA tubes and maintained on ice until plasma isolation within 30 minutes of blood draw. Plasma was isolated by centrifugation at 3000 RPM for 10 minutes at 4°C. Plasma samples (39) were stored at approximately -80°C until shipped to BRT (received on 17 Sep 14) on dry ice. Monkeys were rested for a period of approximately 4 weeks and then administered Treatment #2 (50 mg/kg) on 25 Sep 14. Whole blood samples were obtained by femoral vein puncture from all monkeys at pre-dose (0 hour), 0.5, 3, 6, 12, 24, 48, 72, 120, 180, 240, 360, and 480 hours post-dose (HrPD). Blood was collected into chilled K_2EDTA tubes and maintained on ice until plasma isolation within 30 minutes of blood draw. Plasma was isolated by centrifugation at 3000 RPM for 10 minutes at 4°C. Plasma samples (39) were stored at approximately -80°C until shipped to BRT (received on 16 Oct 14) on dry ice. Plasma samples (78 samples total, Table 2) were logged into inventory at BRT and stored at ≤-70°C until analyzed for LT3114 concentration. Samples were returned to ≤-70°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 2: Monkey plasma PK samples received at BRT.

Animal #	Collection Date	Time Point	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
1	27Aug14	0	1	10	2	5
1	27Aug14	0.5	1	10	2	5
1	27Aug14	3	1	10	2	5
1	27Aug14	6	1	10	2	5
1	27Aug14	12	1	10	2	5
1	28Aug14	24	1	10	2	5
1	29Aug14	48	1	10	2	5
1	30Aug14	72	1	10	2	5
1	01Sep14	120	1	10	2	5
1	03Sep14	180	1	10	2	5
1	06Sep14	240	1	10	2	5
1	11Sep14	360	1	10	2	5
1	16Sep14	480	1	10	2	5
2	27Aug14	0	1	10	2	5
2	27Aug14	0.5	1	10	2	5
2	27Aug14	3	1	10	2	5
2	27Aug14	6	1	10	2	5
2	27Aug14	12	1	10	2	5
2	28Aug14	24	1	10	2	5
2	29Aug14	48	1	10	2	5
2	30Aug14	72	1	10	2	5
2	01Sep14	120	1	10	2	5
2	03Sep14	180	1	10	2	5
2	06Sep14	240	1	10	2	5
2	11Sep14	360	1	10	2	5
2	16Sep14	480	1	10	2	5

Animal #	Collection Date	Time Point	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
3	27Aug14	0	1	10	2	5
3	27Aug14	0.5	1	10	2	5
3	27Aug14	3	1	10	2	5
3	27Aug14	6	1	10	2	5
3	27Aug14	12	1	10	2	5
3	28Aug14	24	1	10	2	5
3	29Aug14	48	1	10	2	5
3	30Aug14	72	1	10	2	5
3	01Sep14	120	1	10	2	5
3	03Sep14	180	1	10	2	5
3	06Sep14	240	1	10	2	5
3	11Sep14	360	1	10	2	5
3	16Sep14	480	1	10	2	5
1	25Sep14	0	2	50	10	5
1	25Sep14	0.5	2	50	10	5
1	25Sep14	3	2	50	10	5
1	25Sep14	6	2	50	10	5
1	25Sep14	12	2	50	10	5
1	26Sep14	24	2	50	10	5
1	27Sep14	48	2	50	10	5
1	28Sep14	72	2	50	10	5
1	30Sep14	120	2	50	10	5
1	02Oct14	180	2	50	10	5
1	05Oct14	240	2	50	10	5
1	10Oct14	360	2	50	10	5
1	15Oct14	480	2	50	10	5
2	25Sep14	0	2	50	10	5
2	25Sep14	0.5	2	50	10	5
2	25Sep14	3	2	50	10	5
2	25Sep14	6	2	50	10	5
2	25Sep14	12	2	50	10	5
2	26Sep14	24	2	50	10	5
2	27Sep14	48	2	50	10	5
2	28Sep14	72	2	50	10	5
2	30Sep14	120	2	50	10	5
2	02Oct14	180	2	50	10	5
2	05Oct14	240	2	50	10	5
2	10Oct14	360	2	50	10	5
2	15Oct14	480	2	50	10	5
3	25Sep14	0	2	50	10	5
3	25Sep14	0.5	2	50	10	5
3	25Sep14	3	2	50	10	5
3	25Sep14	6	2	50	10	5
3	25Sep14	12	2	50	10	5
3	26Sep14	24	2	50	10	5
3	27Sep14	48	2	50	10	5
3	28Sep14	72	2	50	10	5
3	30Sep14	120	2	50	10	5
3	02Oct14	180	2	50	10	5
3	05Oct14	240	2	50	10	5

Animal #	Collection Date	Time Point	Treatment #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
3	10Oct14	360	2	50	10	5
3	15Oct14	480	2	50	10	5

METHODOLOGY

Dose Formulation and Plasma PK Samples

Dose formulation samples (Treatment #1 – 2 mg/mL and Treatment #2 – 10 mg/mL * 3 aliquots each = 6 total samples) were received frozen on dry ice on 17 Sep 14 (2 mg/mL) and 16 Oct 14 (10 mg/mL). Upon receipt at BRT, all samples were stored at ≤-70°C. The samples were analyzed for LT3114 concentrations at BRT.

Monkey plasma PK samples (13 time points * 3 animals/group/time point * 2 treatment groups = 78 total samples) were received frozen on dry ice on 17 Sep 14 (Treatment #1, 10 mg/kg) and 16 Oct 14 (Treatment #2, 50 mg/kg). Upon receipt at BRT, all samples were stored at ≤-70°C. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples and plasma PK samples were analyzed for LT3114 concentration using an ELISA method that was transferred to BRT from the Sponsor (BRT 20140420). The ELISA method is provided in Appendix 1.

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, dilipidized bovine serum (DBS) assay buffer, dose formulation samples, and plasma PK samples were added to the coated plates and incubated to allow for capture of the LT3114. A horse radish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data Analysis

Data analysis was performed using SoftMax®Pro (Molecular Devices) software. Calibration curves were generated by plotting the mean OD_{450} values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD_{450} values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from replicate wells, by the sample dilution factor. Results were reported to one decimal place.

Acceptance Criteria

Calibration Curve

- Determined by the total number of points (with no more than 2 points masked, and no 2 points from the same standard masked).
- Correlation Coefficient ≥ 0.9
- Precision (%CV) between replicates $\leq 20\%$
- Accuracy of each standard within 20% of nominal from 3.9-500 ng/mL.

Sample Acceptance

- Precision (%CV) between replicates $\leq 20\%$
- Accuracy within 20% of nominal (Dose formulation samples only)
- Samples with absorbance values outside the upper limit of quantification (ULOQ at 500 ng/mL) were retested at a higher dilution.
- Samples with absorbance values outside the lower limit of quantification (LLOQ at 5 ng/mL) were retested at a lower dilution, unless already tested at the minimum dilution (1:10).

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples of each dose formulation are provided in Table 3. The concentrations of LT3114 in the 2 mg/mL dose formulations triplicate samples were 2.1, 2.3, and 2.3 mg/mL with a mean LT3114 concentration of 2.2 mg/mL. The concentrations of LT3114 in the 10 mg/mL dose formulation triplicate samples were 10.7, 9.7, and 10.2 mg/mL with a mean LT3114 concentration of 10.2 mg/mL. All individual assessments (aliquots) and the mean LT3114 concentrations for each dose level met acceptance criteria for accuracy and precision.

Monkey Plasma Sample Analysis

Monkey plasma PK samples were analyzed for LT3114 concentrations (Table 4). LT3114 was not detectable in the plasma PK samples collection from monkeys 1 and 3 prior to administration of Treatment #1 (10 mg/kg). The sample collected from monkey 2 prior to administration of Treatment #1 showed an apparent concentration of 0.039 μg/mL of LT3114 which is consistent with minimal matrix interference as observed at the MRD (1:10 dilution) during method transfer (BRT 20140420). LT3114 was detectable in all post-dose plasma PK samples collected following administration of Treatment #1 and concentrations ranged from 9.1 μg/mL to 321.7 μg/mL.

LT3114 was detectable in the Treatment #2 (50 mg/kg) pre-dose plasma PK samples from monkeys 2 (6.5 μg/mL) and 3 (5.9 μg/mL), but was not detectable in the pre-dose plasma PK sample from monkey 1. LT3114 was detectable in all post-dose plasma PK samples from monkeys 2 and 3 throughout the PK time course. However, LT3114 concentration in the plasma of monkey 1 dropped to 0.042 μg/mL at 180 hours post-dose and was undetectable in samples obtained at 240, 360, and 480 hours post-dose. LT3114 concentrations in the plasma PK samples following Treatment #2 ranged from 0.042 μg/mL to 1613.5 μg/mL.

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 1 year from the date that the final report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

There were no deviations from the Study Protocol during the study phase.

TABLES

Table 3: LT3114 concentrations in dose formulation samples.

Dose Formulation	Collection Date	LT3114 (mg/ml)	% of Target	Mean LT3114 (mg/ml)	% of Target	CV%
2 mg/mL (1/3)	27 Aug 14	2.1	105	2.2	110	5.2
2 mg/mL (1/3)	27 Aug 14	2.3	115			
2 mg/mL (1/3)	27 Aug 14	2.3	115			
10 mg/mL (1/3)	24 Sep 14	10.7	107	10.2	102	4.9
10 mg/mL (1/3)	24 Sep 14	9.7	97			
10 mg/mL (1/3)	24 Sep 14	10.2	102			

Table 4: LT3114 concentrations in monkey plasma PK samples.

Collection Timepoint (HrPD)	Date of Collection	Animal Number	LT3114 (µg/ml)
Treatment #1 (10 mg/kg)			
0	27 Aug 14	1	BLQ
		2	0.039
		3	BLQ
0.5	27 Aug 14	1	233.5
		2	265.2
		3	321.7
3	27 Aug 14	1	228.5
		2	214.6
		3	239.6
6	27 Aug 14	1	217.6
		2	216.6
		3	220.5
12	27 Aug 14	1	194.6
		2	172.8
		3	195.1
24	28 Aug 14	1	151.0
		2	156.9
		3	163.0
48	29 Aug 14	1	156.5
		2	151.8
		3	127.3
72	30 Aug 14	1	126.9
		2	124.5
		3	102.7
120	01 Sep 14	1	133.5
		2	108.6
		3	100.6
180	03 Sep 14	1	107.8
		2	96.8
		3	100.9
240	06 Sep 14	1	78.4
		2	83.3
		3	75.1
360	11 Sep 14	1	60.1
		2	68.3
		3	60.0
480	16 Sep 14	1	9.1
		2	50.1
		3	48.7
Treatment #2 (50 mg/kg)			
0	25 Sep 14	1	BLQ
		2	6.5
		3	5.9
0.5	25 Sep 14	1	1244.9
		2	1340.0
		3	1472.2
3	25 Sep 14	1	1034.9
		2	1613.5
		3	1208.0

Collection Timepoint (HrPD)	Date of Collection	Animal Number	LT3114 (µg/ml)
6	25 Sep 14	1	1004.8
		2	1214.0
		3	1016.0
12	25 Sep 14	1	778.4
		2	1133.6
		3	1143.5
24	26 Sep 14	1	687.0
		2	1010.6
		3	1013.3
48	27 Sep 14	1	422.2
		2	716.2
		3	767.8
72	28 Sep 14	1	479.1
		2	333.1
		3	620.5
120	30 Sep 14	1	24.6
		2	611.9
		3	559.5
180	02 Oct 14	1	0.042
		2	591.0
		3	534.6
240	05 Oct 14	1	BLQ
		2	497.8
		3	500.6
360	10 Oct 14	1	BLQ
		2	417.6
		3	409.5
480	15 Oct 14	1	BLQ
		2	300.0
		3	321.5

BLQ – Below Lower Limit of Quantification (3.9 ng/mL)

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Monkey Plasma
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS) (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H_2SO_4 (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)

   1.1 LPA-Conjugated-BSA coating material should be keep frozen until use.

   1.2 After dilution, LPA-Conjugated-BSA coating material should be stored at 2-8° C for up to 4 months

2. Peroxidase conjugated-anti-human IgG secondary antibody

   2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%)

   2.2 Store at -20° C for up to 1 year

3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)

   3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly

   3.2 Store at 2-8°C for up to 3 months

4. Blocking Solution 1xPBS, 1% BSA l, 0.1% Tween 20

   4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X)

   4.2 Mix thoroughly and sterile filter

   4.3 Store at 2-8°C for up to 1 month

ELISA Method

1. Coating Plates

   1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at ≤-70°C. Thaw the aliquot at room temperature.

   1.2 Dilute LPA-Conjugated-BSA coating material to 1 µg/ml in 0.05 M Carbonate Buffer.

   1.3 Coat a 96-well high-binding microtiter plate(s) with 100 µL/well coating solution.

   1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.

2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).

3. Blocking Plates

   3.1 Wash coated plate(s) 4 times with wash solution.

   3.2 Dispense 150 µL/well of blocking solution to each well and cover the plate(s) with plate sealers.

   3.3 Incubate plate(s) at 37°C for 1 hour ± 5 minutes.

4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)

   4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)

   4.2 Prepare unknown samples by diluting in DBS diluent buffer to appropriate dilutions.

   4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS diluent buffer to achieve desired range (0 – 4,000 ng/ml)

   4.4 Wash plate(s) 4 times (300µl/well) with wash solution.

   4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.

5 Add samples to appropriate wells

5.1 Dispense 100 μl of appropriate sample to designated wells in duplicate. Add 100 μl of Diluent Buffer to assay blank wells.

5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.

6 Add HRP-goat-anti-human secondary antibody solution.

6.1 Wash plate(s) 4 times (300μl/well) with wash solution.

6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody

6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.

6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.

6.3 Dispense 100 μl of diluted HRP-goat-anti-human secondary antibody to each well.

6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.

7 Substrate Addition

7.1 Wash coated plate(s) 4 times with wash solution.

7.2 Dispense 100 μl **COLD** TMB solution to each well.

7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.

8 Stop Solution Addition

8.1 Dispense 100 μl 1.0 M H_2SO_4 solution to each well.

8.2 Measure absorbance at 450 nm on microplate reader